

What /pol/ Actually Cites

A citation-level audit of 278,706 posts on 4chan’s
Politically Incorrect board

Marc Garnier*

September 6, 2026

Abstract

4chan’s /pol/ is routinely described as an *alternative media ecosystem*: a space where legacy outlets are rejected and a partisan counter-canon takes their place. The claim is widely repeated and rarely measured. We measure it over 24 daily captures spanning 45 days (2026-07-22 to 2026-09-04): 6,532 threads, 278,706 posts, and 10,078 citation events across 1,408 unique domains, each domain coded against a published codebook.

Three findings work against the framing. First, news outlets of any kind occupy **12.8%** of citation events; social platforms take 43.7% and archives and file hosts 11.0%, the latter cited more often than any single news category. Second, sourcing is extremely concentrated, with a Gini coefficient of 0.821, YouTube alone supplying 28.9% of citations, twelve domains supplying half, and 70.2% of domains appearing exactly once. Third, and most directly, there is no alternative-media cluster in the co-citation structure. Within the news subgraph, assortativity by orientation is -0.042 ; alternative outlets show no tendency to co-occur with one another (1 observed edge against 0.81 expected under a degree-preserving null, $p = 0.62$) and are peripheral (mean degree 0.9 against 2.2 for mainstream). Communities partition by topic and platform affordance, not by editorial orientation.

We report two methodological results alongside. Curated source lists derived from US-centric rating databases undercounted news on this corpus by a factor of 2.8, systematically dropping British, Australian, Israeli and Ukrainian outlets from a corpus dominated by threads about wars outside the United States. And repeated captures of the same threads put a measured floor on survivorship bias: 0.68% of posts (95% CI [0.44, 1.07]) disappear between observations.

We deliberately exclude sentiment analysis. A transformer scored every cited post, but the measure could not be validated on this domain and we judged it unfit to carry a conclusion; Section 3.4 states the reasoning in full.

1 Introduction

4chan’s /pol/ (“Politically Incorrect”) board occupies an outsized place in accounts of online radicalisation and information disorder. Measurement work has characterised its language, its raiding behaviour, and its position in the wider web ecosystem (Hine et al., 2017; Papasavva et al., 2020). A recurring assumption in that literature, and in journalism drawing on it, is that /pol/ constitutes an alternative media ecosystem, with a counter-canon of partisan publishers displacing the legacy press.

*Correspondence: garniermarc.pro@gmail.com. Code, codebook and derived data: <https://github.com/marcgarnier/4chan-pol-source-audit>.

Testing the claim requires knowing what /pol/ cites, in what proportion, and whether its sources organise along a partisan axis. This paper supplies that measurement for a bounded window. Our contributions:

1. **A citation-level audit** of 10,078 citation events with proportions as Wilson score intervals, showing news of any kind at 12.8%.
2. **A concentration analysis** quantifying how narrow the diet is.
3. **A structural test** of the counter-canon claim: co-citation assortativity by editorial orientation, which finds none.
4. **A published codebook** of 203 manually coded domains, and evidence that US-centric source taxonomies undercount news 2.8-fold here.
5. **A measured floor on survivorship bias** from repeated captures.

2 Theoretical framework and prior literature

2.1 Fringe platforms as objects of political measurement

Research on /pol/ began as a measurement problem. Hine et al. (2017) assembled roughly 8M posts over two and a half months and produced the first systematic characterisation of the board, covering posting dynamics, the prevalence of hate speech (12% of /pol/ posts contained a term from a crowdsourced hate lexicon, against 2.2% of a comparison sample of tweets), and the board’s role in coordinating harassment campaigns against other platforms. Papasavva et al. (2020) extended the empirical base to 134.5M posts across 3.5 years, adding toxicity scores and named-entity annotations, and establishing dataset conventions that much later work inherits. They report that 37% of posts carry a Perspective TOXICITY score above 0.5 and that threads average 39.6 posts.

Both studies treat the board as politically significant for what it produces, meaning memes, vocabulary and coordinated aggression, and for how it behaves, meaning velocity, ephemerality and raiding. Neither adopts the citation event as a unit of analysis, and neither reports category-level proportions for the external sources the board circulates. That gap matters, because claims made about /pol/ in political communication typically concern its information supply rather than its output.

Hine et al. (2017) come closest. Their media analysis ranks external domains posted to /pol/ and finds YouTube an order of magnitude ahead of the next two sites, Wikipedia and Twitter, followed by Archive.is, Wikileaks, pastebin and DonaldJTrump.com, with the Daily Mail and Breitbart the first news outlets to appear. They read the absence of CNN, the BBC and the Guardian from the leading ranks as an indication of how polarised opinion on the board has become. That inference rests on rank position within an undifferentiated domain list, a quantity sensitive to how many non-news domains crowd the upper ranks, and it leaves the share of citations going to established outlets distinct from the rank any single outlet occupies. Our design measures the same underlying structure at the level of source categories. Nearly a decade separates the two observation windows, so the comparison we draw in Section 4.3 complements their reading rather than displacing it.

2.2 The alternative media ecosystem thesis

The claim this paper tests is broader than any single study. Fringe political communities are widely held to have constructed an alternative epistemic infrastructure, a counter-canon of partisan

publishers that displaces the established press and supplies the community with its own account of events. The thesis has an appealing internal logic. A community hostile to professional journalism should be expected to substitute something for it, and the outlets named in that substitution are usually the ones whose partisan alignment matches the community's.

Stated carefully, the thesis carries three observable implications for citation behaviour. Alternative outlets should be numerically substantial within the community's information supply. They should travel together, appearing in the same discussions as one another rather than alongside the established press. And they should be structurally central, functioning as reference points that other sources cluster around. Each implication is measurable from citation data alone, without recourse to any model of what participants think.

2.3 Circulation, platforms and the archive

A parallel literature treats fringe communities as nodes in a cross-platform circulation system. Zannettou et al. (2018) trace the propagation of images and links between fringe communities such as 4chan and Gab and mainstream networks such as Twitter and Reddit, establishing that the relevant object of study is the movement of artefacts across platform boundaries rather than the editorial content of any one source. Zannettou et al. (2019) survey the measurement of rumours, hoaxes and unreliable sourcing within online communities, and situate source-sharing as one observable trace of disinformation dynamics.

This literature suggests a rival account of what organises a fringe community's information supply. Circulation may follow platform affordance, meaning what a given host makes easy to embed, mirror or preserve, together with the practical demands of a subject matter, in place of the editorial provenance of sources. Two features of /pol/ make the rival account plausible in advance. Ephemerality is designed into the medium, since threads fall off the catalogue and are permanently deleted after seven days (Papasavva et al., 2020), which gives participants a standing incentive to mirror and archive. And the board's dominant subject matter during our window is armed conflict, a domain in which primary-source tooling substitutes for reporting.

The second feature connects to crisis informatics. Starbird et al. (2014) examine how participants in online discussion assemble accounts of unfolding events from primary sources, live tracking tools and archived material, a practice that operates largely outside the institutional press. Live war mapping, vessel tracking and flight tracking belong to that repertoire, and any audit that codes sources only as news or not-news will record them as residue.

2.4 Source taxonomies as instruments

A final literature bears on the instrument rather than the object. Bozarth et al. (2020) demonstrate that substantive conclusions about the circulation of fake news depend materially on which groundtruth labelling scheme a researcher adopts, comparing AllSides, Ad Fontes and Media Bias/Fact Check and showing that the choice shifts findings. Robertson et al. (2018) rely on comparable third-party labels when auditing partisan bias in search results, which illustrates how widely the practice has been adopted.

We take the point one step further. The rating databases that anchor this practice are built around the American media market, and their coverage thins sharply outside it. On a corpus whose dominant subject is conflict outside the United States, that limitation becomes a measurement failure of the first order, with consequences we quantify in Section 3.3.

Threads captured	6,532
Posts	278,706
Posts with ≥ 1 external link	7,107 (2.55%)
Citation events	10,078
Unique domains	1,408
Daily captures	24
Window	2026-07-22 – 2026-09-04

Table 1: Corpus summary.

2.5 Hypotheses

The framework yields three propositions, each stated so that it can fail.

H1 (counter-canon). If /pol/ maintains an alternative media ecosystem, alternative outlets will be numerically substantial, mutually co-cited and structurally central in the citation network. Sections 4.1 and 4.3 test all three implications.

H2 (infrastructure). If circulation is organised by platform affordance and preservation need, platform and archive hosts will dominate the citation supply, and co-citation communities will partition by subject matter in place of editorial orientation. Sections 4.1 and 4.3 test both parts.

H3 (instrument dependence). If source taxonomies built for the American media market are applied to an internationally sourced corpus, measured news share will be biased downward by an amount large enough to alter substantive conclusions. Section 3.3 quantifies it.

H1 and H2 are rivals over the same evidence, which is what makes the test informative. H3 conditions both, since any estimate of news share is a joint product of the board’s behaviour and the taxonomy applied to it.

3 Data and methods

3.1 Collection

Threads were captured from the live /pol/ JSON API via the 4TCT collector on a daily cadence, giving 24 daily captures between 2026-07-22 and 2026-09-04. This is a snapshot design: conclusions are scoped to the collection window, not to /pol/ in general or across time.

3.2 Citation extraction

The unit of analysis is the **citation event**: one external domain in one post, deduplicated within the post. Links internal to 4chan are excluded. Mobile and shortened hosts normalise to a canonical domain (`youtu.be`, `m.youtube.com` → `youtube.com`), and subdomain matching resolves cases such as `en.wikipedia.org`.

We flag one failure mode because it is a plausible silent defect in any such pipeline. An earlier version of ours retained only the first domain per post. Because /pol/ posts characteristically lead with a video or tweet and append a source, this systematically discarded the appended source: alternative-media citations were undercounted 3.6-fold. All figures here use per-citation counting.

3.3 Source coding and the undercount it corrected

Domains are coded on a medium-type axis into seven categories, namely mainstream, alternative and state-controlled news, social platform, institutional reference, archive/file host, and unclassified. Deterministic structural rules handle platforms, .gov/.mil/.edu suffixes and the Wikimedia family.

An initial pass using curated news lists derived from US-centric rating databases left 47.2% of citations unclassified and put news outlets at 4.6%. That figure was an artefact. The unclassified bucket contained `dailymail.com`, `abc.net.au`, `independent.co.uk`, `itv.com`, `standard.co.uk`, `telegraph.co.uk`, `timesofisrael.com`, `aljazeera.com` and `pravda.com.ua`: British, Australian, Israeli and Ukrainian outlets absent from lists built around the American media market, on a corpus dominated by threads about ongoing wars.

We therefore hand-coded the 200 most-cited unclassified domains, covering 74.9% of that bucket, and publish the result as a versioned codebook of 203 domains. Four rules were fixed in advance and applied uniformly:

1. **Aggregators** (`msn.com`, `yahoo.com`) are coded by what they redistribute; search engines and link shorteners are not news.
2. **State-controlled** is defined by ownership and funding, not editorial independence. This groups Al Jazeera with the BBC, ABC (Australia), PBS, NHK, Deutsche Welle, Anadolu, Channel News Asia and Minnesota Public Radio, and moves the BBC (135 citations) out of mainstream.
3. **Opinion Substacks and blogs** count as alternative media.
4. **Campaign sites** stay unclassified; state media outlets remain state-controlled.

Manual coding raised news media from 4.6% to **12.8%** of citations, a 2.8-fold undercount by the automated pass, and cut the unclassified bucket from 47.2% to 23.6%.

3.4 Why this paper contains no sentiment analysis

We scored every cited post with `cardiffnlp/twitter-roberta-base-sentiment-latest` (Barbieri et al., 2020; Loureiro et al., 2022) and then excluded the results. The reasoning is worth stating, because the measure is a standard one and its omission is deliberate.

Three problems compound. **Construct**: the score is computed over the whole post, so it captures the ambient tone of a message, not the author’s stance toward the source cited in it. Inspection of scored posts finds hostility directed at reported events, at ethnic groups, and at other posters, all of it attributed to whichever domain the post happened to link, including cases where the source is cited approvingly as supporting evidence. **Domain shift**: the model was trained on Twitter. `/pol/`’s irony reads literally (the most positively-scored news post in the corpus is a sarcastic caption on a census map, scored +0.90), and the slurs the board uses as ordinary register are read as hostility regardless of the poster’s actual attitude. **Truncation**: scoring caps at 128 tokens, and 42.3% of cited posts exceed it.

Any of these could be quantified with a hand-annotated validation sample, which we have not produced. Publishing a headline conclusion on an uncalibrated instrument seemed worse than publishing a narrower paper without one. The scoring code remains in the repository behind a flag so the analysis can be revisited after validation.

Everything reported below is derived from citation structure alone.

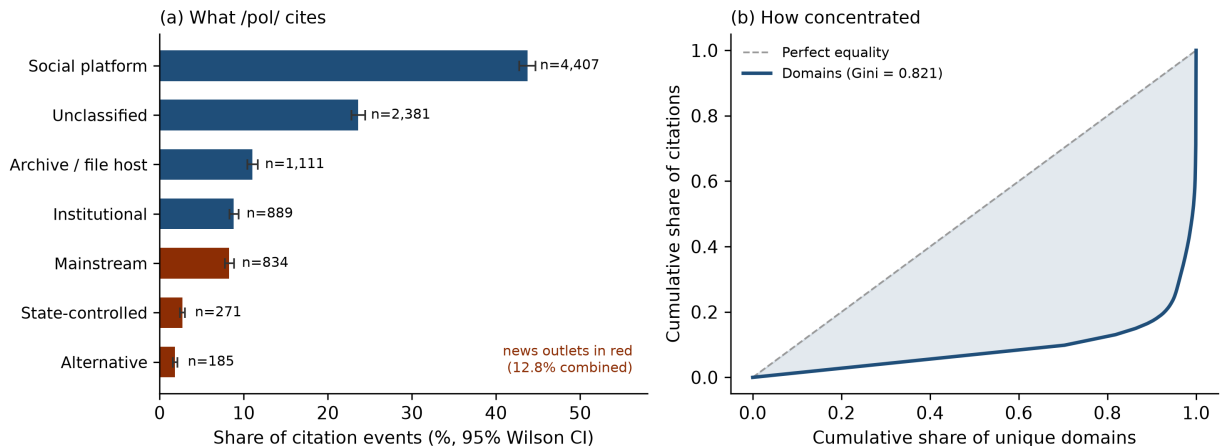


Figure 1: (a) Share of citation events by category with 95% Wilson intervals; news categories in red. (b) Lorenz curve of the citation distribution over 1,408 domains.

Category	n	Share	95% CI
Social platform	4,407	43.7%	[42.8, 44.7]
Unclassified	2,381	23.6%	[22.8, 24.5]
Archive / file host	1,111	11.0%	[10.4, 11.7]
Institutional	889	8.8%	[8.3, 9.4]
Mainstream	834	8.3%	[7.8, 8.8]
State-controlled	271	2.7%	[2.4, 3.0]
Alternative	185	1.8%	[1.6, 2.1]
<i>All news media</i>	<i>1,290</i>	<i>12.8%</i>	<i>[12.2, 13.5]</i>

Table 2: Citation events by category. The rule separates news (below) from non-news (above).

3.5 Statistical analysis

Proportions carry 95% Wilson score intervals (Wilson, 1927). Concentration uses the Gini coefficient and a normalised Herfindahl–Hirschman index over per-domain citation counts. The co-citation graph links domains appearing in the same thread, keeping edges of weight ≥ 2 ; communities come from Louvain, and orientation sorting is measured by attribute assortativity (Newman, 2003). Because alternative domains are rarely cited and therefore sparsely connected, the clustering test permutes labels only among nodes of comparable degree, since a naive label permutation would credit them with connectivity they do not have.

4 Results

4.1 What /pol/ cites

News media account for 12.8% of citation events; platforms for 43.7%; archives and file hosts for 11.0%, more than any single news category. Only 2.55% of posts carry an external link at all.

Proportions are stable across the window with no trend: over the 24 captures, mainstream share ranges 3.6–12.8% (median 8.7%), platform share 37.2–53.7% (median 43.3%) and archive share 6.2–14.4% (median 11.0%).

4.2 Concentration

The diet is narrow. The domain distribution has Gini 0.821 and normalised HHI 0.091. YouTube alone supplies **28.9%** of citations; the top ten supply 49.1% and the top fifty 69.0%. Twelve domains account for half of all citations, while 989 of 1,408 domains (70.2%) appear exactly once (Figure 1b).

4.3 No alternative-media cluster

If /pol/ maintained a counter-canon, alternative outlets should travel together: cited in the same threads as one another rather than alongside the established press. They do not.

The co-citation graph has 286 nodes and 1,846 edges (density 0.045), with Louvain modularity 0.429 over six communities. Assortativity by category over the full graph is +0.054, so domains co-occur with little regard to what kind of source they are.

Restricting to news domains sharpens the test. The news subgraph has 100 nodes (68 mainstream, 20 alternative, 12 state-controlled) and 97 edges. Assortativity by orientation is **-0.042**: if anything slightly disassortative, and indistinguishable from no sorting. Alternative outlets share exactly one edge with another alternative outlet, against 0.81 expected under a degree-preserving null ($p = 0.62$). They are also peripheral: mean degree 0.9, against 2.2 for mainstream and 2.1 for state-controlled. Most alternative citations appear in threads where no other news source is cited at all.

The communities that do exist are topical. The largest (198 domains) is general-purpose, anchored on YouTube, Wikipedia, the BBC and the Guardian, and holds 59 mainstream, 14 alternative and 10 state-controlled domains *together in the same community*. The smaller ones are a war-tracking cluster (`marinetraffic`, `flightradar24`, `flightaware`, `oilprice`), a US-politics platform cluster (`twitter`, `truthsocial`, `rumble`, `tiktok`), and a file-sharing cluster (`files.catbox.moe`, `t.me`, `archive.today`). Sourcing organises by subject matter and platform affordance, not by editorial orientation.

4.4 Archives and OSINT trackers

Two behaviours emerge that a news-centric scheme would hide. Archives and file hosts (`files.catbox.moe`, `reentry.org`, `archive.4plebs.org`, `archive.today`) take 11.0% of citations: hosting media outside platform moderation, and preserving content, including 4chan’s own deleted threads via 4plebs, against removal.

Within the unclassified remainder, 571 citations (5.7% of the corpus, 24.0% of that bucket) across 14 domains are live war-mapping and vehicle-tracking services, namely `liveuamap`, `deepstatemap.live`, `marinetraffic`, `flightradar24`, `adsbexchange`. On a board whose largest identifiable thread topic is armed conflict, primary-source OSINT tooling is cited about half as often as all news media combined.

4.5 A measured floor on survivorship

Sixty-two pairs of repeated captures of the same thread let us measure what is usually asserted: **19 of 2,776 observed posts (0.68%, 95% CI [0.44, 1.07])** had disappeared by the following capture, affecting 7 threads.

This must be read for what it is. It measures post-level deletion *within threads that survived to be captured twice*. It says nothing about threads deleted entirely, nor about threads born and dying between two daily captures, which at a daily cadence on a board with /pol/’s velocity is the larger

loss channel. It is a **floor on content loss** from a self-selected sample, not an estimate of it. We report it because a measured floor is more useful than an unmeasured caveat.

5 Discussion

The counter-canon is not there to be found. On this board, in this window, alternative outlets are neither numerous (1.8% of citations) nor connected to one another (one co-citation edge among twenty domains) nor central (mean degree below one). They appear as isolated citations inside threads organised around topics, next to the mainstream outlets they are supposed to displace. Whatever /pol/’s relationship to the legacy press, “replaced by an alternative ecosystem” does not describe the citation structure.

The frame is also mis-scaled. A literature asking whether /pol/ prefers alternative to mainstream media is asking about 10.1% of what /pol/ circulates. Platforms supply more than three times as many citations as all news combined; archives alone outnumber any single news category. The board’s characteristic act is not passing along an article but passing along a video, a screenshot, a mirror, or a map.

A cautionary note on automated source lists. Our first pass, using curated lists derived from US-centric rating databases, put news at 4.6% of citations. Hand-coding put it at 12.8%. The error was not random: it removed British, Australian, Israeli and Ukrainian outlets from a corpus whose dominant subject is war outside the United States. Any audit applying American media-bias taxonomies to an internationally-sourced corpus should expect an undercount of this order and check for it directly.

What we cannot say. Nothing here speaks to how /pol/ *treats* the sources it cites. Citation is not endorsement, and on a board whose characteristic mode is derision it may often be the opposite. Establishing that would require the stance measurement this paper declines to fake (Section 3.4), and it remains the obvious next study.

6 Limitations

The codebook is single-coder. The 203 domains were coded by one author against fixed written rules, with 12 uncertain cases flagged. No second coder annotated independently, so no Cohen’s κ or Krippendorff’s α is reported. Structural categories (platform, archive, institutional) rest on deterministic rules and are robust; the news subdivisions rest on that single judgment.

The network test is small. Twenty alternative domains and 97 news-subgraph edges is a thin basis. The absence of clustering is consistent with the descriptive picture and with the community structure, but a larger corpus could reveal structure this one cannot resolve. We report it as evidence against the counter-canon claim, not as proof of its impossibility.

Coding is single-axis. A fuller design codes each domain on three orthogonal axes (medium type, editorial orientation, ownership) against external databases, since forcing them into one label is what makes Fox News, BBC and France 24 irreconcilable. We code medium type only, with ownership folded into the state-controlled category by rule. Our orientation test inherits that coarseness:

it asks whether mainstream, alternative and state-controlled outlets cluster, not whether a finer left-right gradient does.

A quarter of citations remain unclassified. 23.6% fall outside all categories, concentrated in a long tail of which 904 domains are cited exactly once. Hand-coding covered 74.9% of the original unclassified bucket; the remainder could contain further news outlets, which would raise the 12.8% figure again.

Survivorship bias is bounded below, not corrected. See Section 4.5. Moderation removes the most extreme content preferentially, so the surviving corpus is filtered in the direction of the phenomenon studied. Source-distribution estimates are relatively robust, since deletion would need to correlate with cited-source category.

Scope. One board, 24 daily captures, 45 days. Results describe the surviving, publicly visible discourse of /pol/ during that window, and do not generalise across time or to other imageboards.

7 Conclusion

/pol/’s information diet is not a media diet. News outlets of every political description together supply an eighth of its citations, while platforms supply more than four in ten and archives more than one in ten. The alternative press that the board is supposed to have adopted in place of the legacy press is rare, peripheral, and unclustered: its outlets do not travel together, they travel beside the mainstream outlets they were supposed to replace, inside threads organised by subject matter.

Two methodological results follow the substantive one. Source taxonomies built for the American media market misdescribe an internationally-sourced corpus badly enough to invert a headline figure. And a measure that cannot be validated on its target domain is better excluded than caveated, since a narrower paper is worth more than a broader one resting on an uncalibrated instrument. The productive next step is the stance measurement this paper declines to attempt: what /pol/ does with the sources it cites, as opposed to which ones it cites.

Data and code availability

Analysis code, the manual domain codebook, derived statistics and figures are at <https://github.com/marcgarnier/4chan-pol-source-audit>. The raw capture corpus (~600 MB) is not redistributed but is reproducible from the public 4chan JSON API using the collection scripts provided. Every number in this paper is regenerated by `analysis.py` from the released database schema and codebook.

References

- Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. *Findings of EMNLP 2020*.
- Bozarth, L., Saraf, A., & Budak, C. (2020). Higher Ground? How Groundtruth Labeling Impacts Our Understanding of Fake News about the 2016 U.S. Presidential Nominees. *ICWSM 2020*.

- Hine, G. E., Onaolapo, J., De Cristofaro, E., Kourtellis, N., Leontiadis, I., Samaras, R., Stringhini, G., & Blackburn, J. (2017). Kek, Cucks, and God Emperor Trump: A Measurement Study of 4chan’s Politically Incorrect Forum and Its Effects on the Web. *Proceedings of the 11th International AAAI Conference on Web and Social Media (ICWSM 2017)*.
- Loureiro, D., Barbieri, F., Neves, L., Espinosa-Anke, L., & Camacho-Collados, J. (2022). TimeLMs: Diachronic Language Models from Twitter. *ACL 2022, System Demonstrations*.
- Newman, M. E. J. (2003). Mixing patterns in networks. *Physical Review E*, 67(2), 026126.
- Papasavva, A., Zannettou, S., De Cristofaro, E., Stringhini, G., & Blackburn, J. (2020). Raiders of the Lost Kek: 3.5 Years of Augmented 4chan Posts from the Politically Incorrect Board. *ICWSM 2020*.
- Robertson, R. E., Jiang, S., Joseph, K., Friedland, L., Lazer, D., & Wilson, C. (2018). Auditing Partisan Audience Bias within Google Search. *CSCW 2018*.
- Starbird, K., Maddock, J., Orand, M., Achterman, P., & Mason, R. M. (2014). Rumors, False Flags, and Digital Vigilantes: Misinformation on Twitter after the 2013 Boston Marathon Bombing. *iConference 2014*.
- Zannettou, S., Caulfield, T., Blackburn, J., De Cristofaro, E., Sirivianos, M., Stringhini, G., & Suarez-Tangil, G. (2018). On the Origin of Memes by Means of Fringe Web Communities. *Proceedings of the ACM Internet Measurement Conference (IMC 2018)*.
- Zannettou, S., Sirivianos, M., Blackburn, J., & Kourtellis, N. (2019). The Web of False Information: Rumors, Fake News, Hoaxes, Clickbait, and Various Other Shenanigans. *ACM Journal of Data and Information Quality*, 11(3), 1–37.
- Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212.